

2026-05-22 - Debugging and Testing Zephyr Applications

Goal

Understand the Zephyr debugging and testing workflow for finding and fixing issues.

What you will learn

- How to use logging and console output for debugging
- How to run Zephyr tests and sample validation
- The role of `west test`, `west debug`, and `west flash`
- How to interpret build and runtime errors

Overview

Debugging Zephyr often involves enabling log output, using debugger tools, and running built-in test suites. These tools help you verify that your code and configuration are correct.

Debug techniques

- `printf()` and `LOG_INF()` for runtime messages
- `west debug` to attach GDB to the target
- `west test` for Zephyr test cases
- `west build -t run` on simulation boards such as `native_posix`

Why this matters

Good debugging practices accelerate learning, reduce frustration, and help you understand how your application interacts with Zephyr internals.

Practice tasks

1. Add logging to an existing Zephyr app and watch the console output.
2. Run a sample on `native_posix` or a simulator with `west build -b native_posix`.
3. Review build warnings and fix one configuration or code issue.

Next topic

After this, you can continue with advanced Zephyr concepts such as Bluetooth, networking, and power management.